Work in progress — draft, not peer reviewed

The Frohmanian Tethered Metriplectic World Model in Lean 4: Canonical Bracket Splitting, Encoding Obstructions, and the Newton Potential

Companion paper to `BenFrohman/metriplectic-world-model`

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Draft — tracks build in progress. This repository formalizes a *model*: the tethered metriplectic system, which adds a symmetric tether bracket and cubic damping of peak vorticity to 3D vorticity dynamics. It proves nothing about unmodified Navier–Stokes ($\kappa = 0$) and does not resolve the Clay Millennium problem.

Abstract

We report a Lean 4/Mathlib [2, 4] formalization (0 `sorry`; axioms only `propext`, `Classical.choice`, `Quot.sound`) of the author's tethered metriplectic model (in the sense of [3]). Proved unconditionally: the unique antisymmetric/symmetric splitting of a bilinear bracket (`split_exists_unique`); a non-circular canonicity theorem forcing the metric bracket to the form $c\langle \Pi_u x, \Pi_u y\rangle$ (`canonical_metric_bracket`); a uniform vorticity ceiling for the comparison inequality $M' \le CM^2 - \kappa M^3$; a log-loss replacement for the stretching hypothesis giving a ceiling when $\kappa > 4C_{\mathrm{abs}}$; several encoding counterexamples; the Newton-potential identity $\Delta(N \star g) = -4\pi g$ on $\mathbb{R}^3$ for $g \in C_c^2$; and $\operatorname{curl} \mathrm{BS}\,\omega = \omega$ for smooth, compactly supported, divergence-free ω (`curlBiotSavartTest`). Conditional and open items are listed by their exact Prop names.

2020 Mathematics Subject Classification. 35Q30, 76D05, 68V15; 37K05, 76B47, 03B35.
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1 Naming

`Pi_u` is the author's own name (manuscript of 31 May 2026): the projection off span$\{u\}$,

$$\Pi_u x = x - \frac{\langle u, x\rangle}{\|u\|^2}\, u. \tag{1.1}$$

All other identifiers defined in this repository are *coined in this build* (marked †), including `canonical_metric_bracket`, kept by the author's choice and flagged as coined. Modules under `FrohmanianSymplecticTether/` and `TetherMayForm/` are ported verbatim from earlier repositories.

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2 Manuscript framing

The May/June 2026 manuscript introduces a tethered Lyapunov functional S_ε and reduces regularity to the comparison ODE

$$y' = Cy^2 - \kappa'' y^3 \tag{2.1}$$

for the peak vorticity, through transport cancellation, viscous dissipation at the maximum, a Hölder–Sobolev gate, and Young absorption at $\varepsilon = \kappa/4$ (Proposition 2). Within the model, the cubic term comes from the tether. The manuscript's hypothesis (H5), a non-circular majorant for the vorticity, is the genuine gap; for $\kappa = 0$ it is the Clay core. Here κ has units of time and breaks Navier–Stokes scaling, which is why the model is distinct from NS. *Normalization of κ:* where the Lean code fixes $\kappa = 3/2$ (`CalderonZygmundConstant3D`, defined as $(3/(8\pi))\cdot 4\pi$), this is a convention; it is not a computed Calderón–Zygmund operator norm and not derived from a Conley–Zehnder index. The ceilings hold for every $\kappa > 0$.

3 Brackets and canonicity (`Bracket.lean`, `Canonical.lean`)

Theorem 3.1. `split_exists_unique`†*: every bilinear bracket is uniquely an antisymmetric part plus a symmetric part.*

Theorem 3.2. `canonical_metric_bracket`†*: on a real inner product space, a bilinear form B that is symmetric (C1), degenerate along u (C2: $B(u,\cdot) = 0$) and invariant under every linear isometry fixing u equals*

$$B(x, y) = c\,\langle \Pi_u x, \Pi_u y\rangle.$$

The hypotheses do not mention Π_u. `canonical_metric_bracket_sign`†*: if $B(x,x) \le 0$ then $c \le 0$.* `Pi_u_form_symm_degen`†*: the forced form satisfies C1, C2.*

Scope: pointwise at one state; does not force the $|\omega|^2$ weight, the value of κ, or C3. Also proved: `energy_conserved`, `energy_const_along`, `entropy_production`, `gen_entropy_eq`, `entropy_monotone_along`†.

4 Ceiling and stretching (`Ceiling.lean`, `AuditFixes.lean`)

Theorem 4.1. `uniform_ceiling`†*: if $M' \le CM^2 - \kappa M^3$ and $\kappa > 0$, then*

$$M(t) \le \max\big(M(0), \max(C/\kappa, 0)\big) \qquad (t \ge 0). \tag{4.1}$$

Also `absorbing`, `ceiling_antitone_kappa`, `kappa_zero_no_ceiling`† *(at $\kappa = 0$, $1/(1-t)$ satisfies $M' \le M^2$ and blows up).*

Correction. The hypothesis `StretchingCZ` (a pure y^3 bound on stretching at the maximum) is *false as a sup-norm bound* for Navier–Stokes: the known bound carries a logarithmic loss. Results using it are relabeled as model-only. Its honest replacement is `StretchingLogLoss` $d\,C_1\,C_2$†:

$$St \le C_1 y^3 + C_2 y^3 \log(1 + y).$$

Theorem 4.2. `maxpoint_rate_logLoss`†*: under* `EnergyIdentity`, `StretchingLogLoss`, `ViscousAtMax`, `TetherFeedback`, `CrossControlled`,

$$y' \le \Big(C_1 + \tfrac{\kappa}{4} + \tfrac{4C_2^2}{\kappa}\Big) y^2 - \Big(\tfrac{\kappa}{4} - C_{\text{abs}}\Big) y^3,$$

so with `uniform_ceiling` *a uniform ceiling follows whenever $\kappa > 4C_{\text{abs}}$ (Young absorption at $\varepsilon = \kappa/4$).*

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5 Encoding obstructions (`ChoiceFree`, `HelmholtzObstruction`, `TestClass`)

- `not_streamFunctionRepModGrad`†: `StreamFunctionRepModGrad` fails for every ω — representatives are determined only a.e.; uniqueness must be stated almost everywhere (open).
- `arnoldBracketOf_antisymm`, `entropyCasimir_arnold`†: a choice-free Arnold bracket [1] and a Casimir criterion.
- `not_curl_biotSavart_eq`†: $\operatorname{curl}(\mathrm{BiotSavart}\,\omega) = \omega$ fails for the point spike in the current encoding; `helicity_casimir_of_curlBiotSavart`† is conditional on `CurlBiotSavartOn`.
- `IsTestOrbit`† (smooth, compactly supported, div-free); `not_isTestOrbit_pointSpike`, `helicity_casimir_test`† (conditional on `CurlBiotSavartTest`, discharged in `helicity_casimir_test_closed`, Section 7); `integrableOn_ball_norm_rpow_neg`†.

6 Newton potential on $\mathbb{R}^3$ (`NewtonKernel`, `RegularizedKernel`, `NewtonLaplacian`)

Proved: `scalarLaplacian_newtonRadial` ($\Delta(1/|y|) = 0$, $y \neq 0$), `contDiff_newtonPotential`, `hasFDerivAt_newtonPotential` (differentiation under the integral); for $N_\varepsilon = (|y|^2 + \varepsilon^2)^{-1/2}$: `scalarLaplacian_regNewton` ($\Delta N_\varepsilon = -3\varepsilon^2(|y|^2 + \varepsilon^2)^{-5/2}$), `regNewton_le`, `tendsto_regNewton`, `integral_neg_scalarLaplacian_regNewton` ($\int -\Delta N_\varepsilon = 4\pi$), `tendsto_regPeak_approxIdentity` (all †). The regularized route avoids spherical Green identities, which Mathlib lacks. Step 3 is closed (`NewtonLaplacian`): `integral_mul_scalarLaplacian_eq` ($\int K\,\Delta g = \int \Delta K\, g$ for $K \in C^2$, $g \in C^2_c$, by Mathlib integration by parts, no boundary terms), `regNewton_conv_laplacian` ($N_\varepsilon \star \Delta g = (\Delta N_\varepsilon) \star g$), `tendsto_regNewton_conv` (the $\varepsilon \to 0$ limit, dominated by $N|h|$), `scalarLaplacian_newtonPotential` ($\Delta(N \star g) = N \star \Delta g$), and

Theorem 6.1. `scalarLaplacian_newtonPotential_eq`†*: for $g \in C^2_c(\mathbb{R}^3)$ and $N(y) = 1/|y|$, $\Delta(N \star g)(x) = -4\pi g(x)$ for every x.*

7 Biot–Savart on the test class (`BiotSavartNewton`)

With $G_j(t) = t_j/|t|^3$ (`gradKer`† at $\varepsilon = 0$):

Theorem 7.1. `newtonPotential_deriv_eq`†*: for $h \in C^1_c$, $N \star \partial_j h = -\int G_j(t)\, h(x - t)\, dt$. (Regularized identity `regNewton_conv_deriv`, then $\varepsilon \to 0$ on both sides; the right side is dominated by $|t|^{-2}|h|$.)*

Theorem 7.2 (Step 5)**.** `biotSavart_eq_curl_newtonVec`†*: for* `IsTestOrbit` *ω,* $\mathrm{BiotSavart}\,\omega = (4\pi)^{-1}\operatorname{curl}(N \star \omega)$ *pointwise, $N \star \omega$ taken componentwise (*`newtonVec`*).*

Theorem 7.3 (Step 4)**.** `curl_curl_newtonVec`†*: for* `IsTestOrbit` *ω,* $\operatorname{curl}\operatorname{curl}(N \star \omega) = 4\pi\omega$*, using symmetry of second derivatives (*`fderiv_fderiv_swap`*),* $\operatorname{div}(N \star \omega) = N \star \operatorname{div}\omega = 0$*, and the step-3 theorem.*

Theorem 7.4. `curlBiotSavartTest`†*:* `CurlBiotSavartTest` *holds, i.e.* $\operatorname{curl}\mathrm{BiotSavart}\,\omega = \omega$ *for every smooth, compactly supported, divergence-free ω. Consequently* `helicity_casimir_test_closed`†*: helicity is a Casimir of the Arnold bracket at such ω, given the explicit argument $D\,S = u$.*

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Scope: this is the classical identity on the smooth compactly supported class only; it says nothing about the dynamics. The Gâteaux derivative of helicity is treated in Section 8.

8 Helicity on the test class (`HelicityGateaux`)

Definition fix. `IsSmoothOrbit` previously used `ContDiff` $\mathbb{R}$ $\top$, which in current Mathlib means analytic; a compactly supported test orbit was then 0 and the earlier test-class statements were vacuous. It now uses C^∞ (the order $(\top : \mathbb{N}_\infty)$ coerced to `WithTop` $\mathbb{N}_\infty$); the class contains every $\operatorname{curl}\psi$ with $\psi \in C_c^\infty$ (`isTestOrbit_curlOrbit`†).

Theorem 8.1. `biotSavart_selfAdjoint`†*: for continuous compactly supported* a, b, $\int\langle Ba, b\rangle = \int\langle Bb, a\rangle$ *(Fubini with the majorant* $|x-y|^{-2}|a(y)||b(x)|$*, kernel symmetry, triple product).*

Theorem 8.2. `helicity_gateaux`†*: with* $\mathcal{H}(\omega) = \int\langle B\omega, \omega\rangle$ *(*`helicity`†*),* $2B\omega$ *is a Gâteaux representative of* $\mathcal{H}$ *on the test class.*

Theorem 8.3. `curl_eq_of_isGateauxRepOn`†*: two* C^1 *representatives of the same functional on the test class have equal curl. Hence* `helicity_casimir_of_rep`†*: if* $D\,S$ *is any* C^1 *representative of* $\mathcal{H}$*, then* $\operatorname{curl} D\,S = 2\omega$ *and* S *is a Casimir of the Arnold bracket* `ArnoldBracketOf`*; no hypothesis on the value of* $D\,S$*.*

Theorem 8.4. `streamFunctionRepModGradOn_of_exists`†*:* `StreamFunctionRepModGradOn` *follows from one* C^2 *kinetic-energy representative* ψ *with* $\operatorname{curl}\psi = u$ *and the Poincaré lemma* `CurlFreeIsGradient`†*; the curl half of uniqueness is* `kineticEnergy_reps_curl_eq`†*.*

9 Poincaré lemma and the kinetic-energy representative (`PoincareR3`, `KineticEnergyRep`)

Theorem 9.1. `curlFreeIsGradient`†*: every* C^2 *curl-free field on* $\mathbb{R}^3$ *is the gradient of a* C^2 *function.*

Theorem 9.2. `decays_newtonPotential`†*: if* g *is continuous, compactly supported and* $\int g = 0$*, then* $|N \star g(x)| \le C(1+|x|)^{-2}$*. Derivatives of* C_c^1 *functions and the coordinates of a divergence-free test field have mean zero (*`IsTestOrbit.integral_coord_eq_zero`†*).*

Theorem 9.3. `integral_pairing_curl_symm`†*:*

$$\int \langle a, \operatorname{curl} b\rangle \,\mathrm{d}x = \int \langle \operatorname{curl} a, b\rangle \,\mathrm{d}x$$

for fields with decaying coordinates and first derivatives. Hence, with $\psi = (4\pi)^{-1} N \star \omega$ *(*`keStream`†*),* $\int\langle B\omega, B\varepsilon\rangle = \int\langle\psi, \varepsilon\rangle$*, so* ψ *is a* C^∞ *Gâteaux representative of the kinetic energy with* $\operatorname{curl}\psi = u$ *(*`keStream_isGateauxRep`†*).*

Theorem 9.4. `streamFunctionRepModGradOn`†*:* `StreamFunctionRepModGradOn` *holds unconditionally on the test class.*

Smoothness audit. `instSmooth` in `Instance.lean` stated analyticity where C^∞ was meant (still true, now stated as C^∞). `contDiffAt_newtonRadial` states analyticity of $1/|x|$ off 0; that is true and proved, so it is left as is.

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10 Maximizer data from a vorticity field (`MaxPointDanskin`, `EntropyCasimirArnold`)

Theorem 10.1. `danskin_upper_dini`†*: if $f, \partial_t f$ are jointly continuous on $\mathbb{R} \times K$, K compact, then for some maximizer $x_\star$ of $f(t,\cdot)$ on K and every $\varepsilon > 0$, $\sup_K f(t+h) \le f(t,x_\star) + h(\partial_t f(t,x_\star) + \varepsilon)$ for small $h > 0$; i.e. $D^+ \sup_K f(t) \le \partial_t f(t,x_\star)$.* `vorticity_danskin_upper_dini`† *applies it to $|\omega|^2$.*

Theorem 10.2. `sup_mul_deriv_eq_inner`†*: if $Y = \sup|\omega|$ is differentiable at t, then $YY' = \omega \cdot \partial_t \omega$ at every global maximizer. At such a point $\omega \cdot (v \cdot \nabla)\omega = 0$ (*`inner_convective_eq_zero_of_max`†*) and $\omega \cdot \Delta\omega \le 0$ for C^2 fields (*`inner_laplacian_nonpos_of_max`†*).*

Theorem 10.3. `maxPointRealization_of_field`†*: for a solution of* `ModelVorticityPDE` *with $\nu \ge 0$, $\omega(t) \in C^2_c$, C^1 in time, and $Y = \sup|\omega| > 0$ differentiable,* `MaxPointRealization` *holds with* `EnergyIdentity`, `ViscousAtMax` *and* `CrossControlled` *discharged. The remaining hypotheses are* `FieldStretchingBound`† *and* `FieldTetherFeedback`†*. Running the ceiling argument from the one-sided Dini bound alone is open.*

Theorem 10.4. `arnoldCasimirField_helicity`†*: $2u$ satisfies* `ArnoldCasimirField`† *($\forall a$,* `ArnoldBracketOf` *$\omega\, a\, (2u) = 0$) on the test class. This is the clean form of* `EntropyCasimir` *for the concrete bracket (*`entropyCasimir_of_arnoldCasimirField`†*). The choice-based form* `EntropyCasimirChoice`† *is kept and marked not provable as stated.*

11 The ceiling from the Dini bound (`DiniCeiling`)

Theorem 11.1. `ceiling_of_upper_dini`†*: if M is continuous on $[0,\infty)$, $M(t+h) \le M(t) + h(R(t) + \varepsilon)$ for small $h > 0$, and $R(t) < 0$ whenever $M(t) > B$, then $M(0) \le B$ implies $M \le B$ on $[0,\infty)$.*

Theorem 11.2. `field_uniform_ceiling`†*: take a solution of* `ModelVorticityPDE` *with $\nu \ge 0$, $\kappa > 0$, C^2 in space, C^1 in time, and support in a fixed compact set. Under* `FieldStretchingBound` *and* `FieldTetherFeedback`*, $\sup|\omega(t)| \le \max(\sup|\omega(0)|, C/\kappa, 0)$ for all $t \ge 0$. No differentiability of $\sup|\omega|$ is assumed; its continuity is proved (*`sup_continuous_of_uniform_support`†*).*

Theorem 11.3. `fieldTetherFeedback_cubicTether`†*: for $T = -\kappa|\omega|^2\omega$ (*`cubicTether`†*),* `FieldTetherFeedback` *holds. Hence* `field_uniform_ceiling_cubicTether`† *needs only* `FieldStretchingBound`*. For a general tether it does not follow.*

12 Log-loss stretching with the cubic tether (`LogLossCeiling`)

Theorem 12.1. `field_uniform_ceiling_logLoss_cubicTether`†*: take a solution of* `ModelVorticityPDE` *with the cubic tether, under the regularity and uniform-support hypotheses of* `field_uniform_ceiling`*. Assume the named Prop* `FieldStretchingLogLoss`†*, $\omega \cdot (\omega \cdot \nabla)u \le C_1 Y^3 + C_2 Y^3 \log(1+Y)$ at maxima. Then $\sup|\omega(t)| \le \mathrm{ceiling}(Y_0, C_1 + \kappa/4 + 4C_2^2/\kappa, 3\kappa/4)$ for every $\kappa > 0$.*

This is a reduction, not an unconditional result. The Calderón–Zygmund estimate is not formalized here. Its known form has $\log^+\big(\|\omega\|_{C^\alpha}/Y\big)$, a higher norm that is not controlled uniformly in time, so it would not close the argument. The Y-only log loss is a strictly stronger assumption.

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13 Status

- **Proved unconditionally:** `split_exists_unique`, `canonical_metric_bracket` (+ sign, symm/degen), energy/entropy structure, `uniform_ceiling`, `maxpoint_rate_logLoss` (as an implication from its listed Props), the obstructions above, the Newton potential identity (steps 1–3), `curlBiotSavartTest` (steps 4–5), Biot–Savart self-adjointness, the Poincaré lemma `curlFreeIsGradient`, `streamFunctionRepModGradOn` on the test class, $D\mathcal{H} = 2u$ and the helicity Casimir of the concrete Arnold bracket on the test class (`helicity_casimir_of_rep`), the scalar instance `instance_global_regularity`, the Danskin bound `danskin_upper_dini` and the maximum-principle lemmas.
- **Conditional on named Props:** `model_uniform_ceiling` (`TetheredMaxRate`); `model_global_regularity` (`TetheredMaxRate`, `ModelContinuation`); `model_global_regularity_of_realization` (`MaxPointRealization`, whose fields include `StretchingCZ` — false as an L^∞ NS bound); velocity-tether energy/entropy (`KineticEnergyCurlIdentity`, `EntropyCasimir`); helicity Casimir on the test class given $D\,S = u$ (`helicity_casimir_test_closed`; no longer conditional on `CurlBiotSavartTest`).
- **Open:** `EntropyCasimir` for the choice-based `FunctionalDerivative` bracket (not provable as stated); deriving the $|\omega|^2$ weight, κ and C3 from C1–C3; for the field-level realization (`maxPointRealization_of_field`): `FieldStretchingBound` (or `FieldStretchingLogLoss`; the CZ estimate is not formalized, and its C^α form would not close the ceiling), `FieldTetherFeedback` for tethers other than `cubicTether`, and existence and regularity (uniform compact support) of actual solutions. Differentiability of $\sup|\omega|$ is no longer needed (`field_uniform_ceiling`); and, for unmodified NS, the (H5) majorant, i.e. the Clay core. Neither the Clay problem nor NS regularity is claimed.
- The type named `T3` is $\mathbb{R}^3$, not the torus.

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